Program 5:

requirements:

int, n1, n2, n3, n4, n5;
double sum, productSqrt, sumRat, harMean, arithMean, powSum, stedv, pow(1-5)

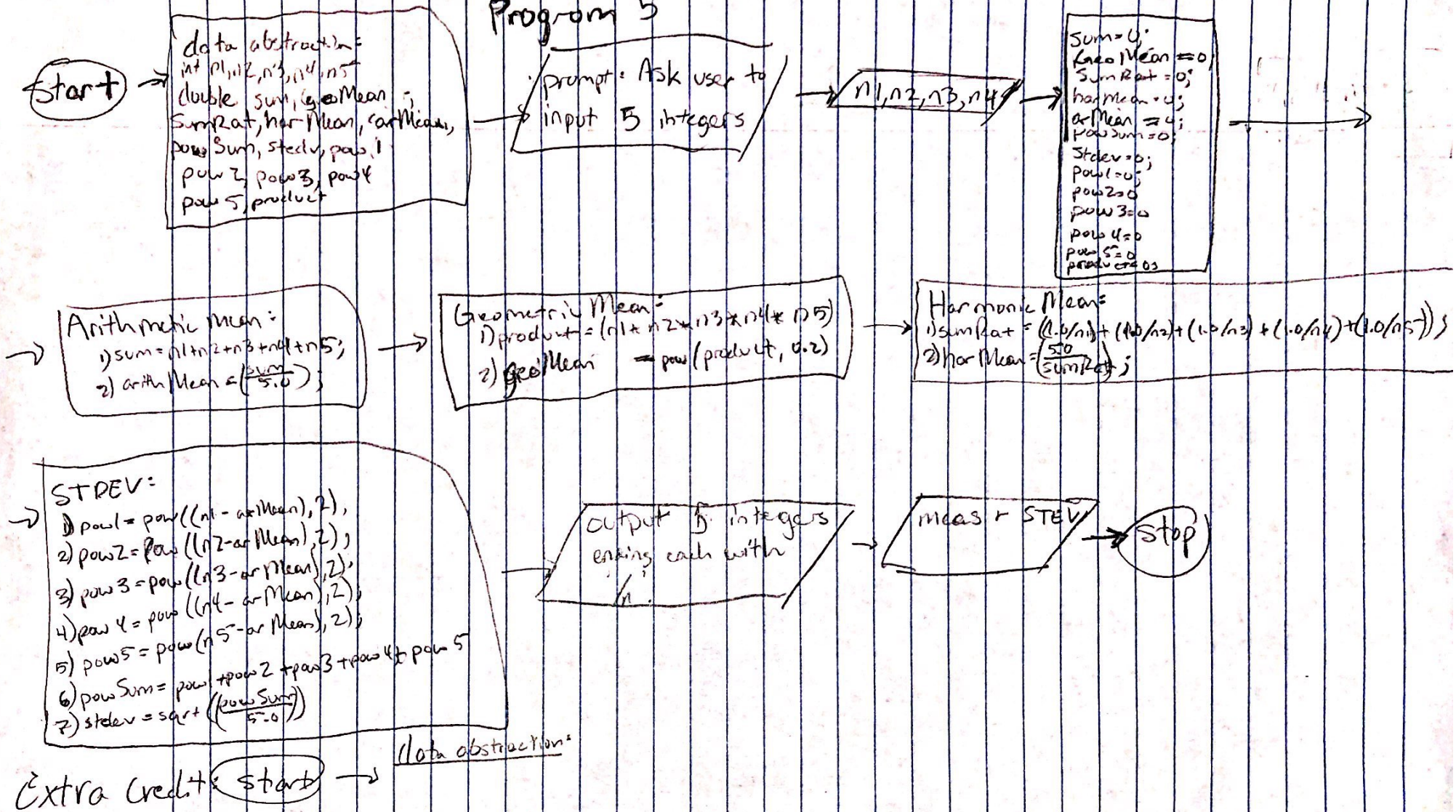
- Arithmetic mean
  1) sum = n1 + n2 + n3 + n4 + n5
  2) arithMean = $\frac{sum}{5.0}$
- Geometric Mean =
  1) product = (n1 * n2 * n3 * n4 * n5)
  2) productSqrt = pow(product, 0.2)
- Harmonic Mean
  1) sumRat = ((1.0/n1) + ($\frac{1.0}{n2}$) + (1.0/n3) + ($\frac{1.0}{n4}$) + ($\frac{1.0}{n5}$))
  2) harMean = $\frac{5.0}{sumRat}$
- STDEV
  1) pow1 → pow((n1 - arMean), 2)
  2) pow2 → pow((n2 - arMean), 2)
  3) pow3 → pow((n3 - arMean), 2)
  4) pow4 → pow((n4 - arMean), 2)
  5) pow5 → pow((n5 - arMean), 2)
  6) powSum = pow1 + pow2 + pow3 + pow4 + pow5
  7) stedv = sqrt($\frac{powSum}{5.0}$)

Extra credit

- Root Mean Square
  1) double squareAve, rootMeanSquare
  2) squareAve = (pow(n1, 2) + pow(n2, 2) + pow(n3, 2) + pow(n4, 2) + pow(n5, 2))
  3) rootMean Square = sqrt(squareAve / 5.0)
- 20% Winsorized mean
  1)

Scanned with CamScanner

Program 5



Extra Credit: Start → data abstraction:

Scanned with CamScanner